Sample Paper 8 — Answer Key (Level-2 Difficulty Step-Up)

Total: 40 marks (30 × 1 mark + 5 × 2 marks), 60 minutes.

Section 1 — Logical Reasoning

1. **D) Circle** — Triangle, Square, and Pentagon are shapes with straight sides and corners; a Circle has neither.
2. **A) School** — A doctor works at a hospital; a teacher works at a school.
3. **A) KYPICR** — The rule is "−2 letters" (wrapping A → Y): M → K, A → Y, R → P, K → I, E → C, T → R.
4. **C) 72** — Two patterns are interleaved: the odd-position terms go 5, 10, 15, 20 (+5 each); the even-position terms go 90, 84, 78, ? (−6 each) — so the next even term is 72.
5. **A) Esha, Bhavna, Chirag, Divya, Arjun** — Chirag is at position 3 (middle). Esha is at the left end, position 1. Divya is immediately right of Chirag, position 4. Bhavna sits between Esha (1) and Chirag (3), so Bhavna is position 2. The only person left, Arjun, takes position 5.
6. **A) Back at the starting point** — South and North cancel (12 m each way); East and West cancel (9 m each way).
7. **B) Tuesday** — 30 − 7 = 23 days later; 23 = 3 weeks + 2 days, so the day of the week moves forward 2 days from Sunday: Tuesday.
8. **C) 24** — 9 = 3×3, 16 = 4×4, 49 = 7×7 (each a number multiplied by itself — a "square number"); 24 does not fit this pattern.
9. **D) 95** — Each term is double the previous term plus 1: 23×2+1 = 47, 47×2+1 = 95.
10. **B) 11th** — Being 15th from the right means 14 students are to Priya's right. The 4 students who leave from the right end are among those to her right, so her position from the right decreases by 4: 15 − 4 = 11th.

Section 2 — Mathematical Reasoning

11. **A) 84443** — 56789 + 27654 = 84443.
12. **A) 41235** — 90000 − 48765 = 41235.
13. **A) 27648** — 3456 × 8 = 27648.
14. **A) 532** — 9576 ÷ 18 = 532 exactly.
15. **C) 12** — 60 = $2^2 \times 3 \times 5$, 84 = $2^2 \times 3 \times 7$. HCF = $2^2 \times 3$ = 12.
16. **B) 36** — 12 = $2^2 \times 3$, 18 = 2×3^2. LCM = $2^2 \times 3^2$ = 36.
17. **C) 2/3** — As decimals: 5/9 ≈ 0.556, 7/12 ≈ 0.583, 2/3 ≈ 0.667, 3/5 = 0.6. The greatest is 2/3.

18. **B) 8 km 150 m** — 4250 m + 3900 m = 8150 m = 8 km 150 m.
19. **C) 1,260 sq m** — Area = 45 × 28 = 1,260 sq m.
20. **C) 12 cm** — A regular octagon has 8 equal sides: 96 ÷ 8 = 12 cm.

Section 3 — Everyday Mathematics

21. **C) ₹18,700** — 93500 ÷ 5 = 18700.
22. **C) 1,120** — Rate = 2520 ÷ 9 = 280 toys/hour; in 4 hours = 280 × 4 = 1,120.
23. **C) 28 min** — 4200 ÷ 150 = 28 minutes.
24. **C) 4 h 45 min** — Total time 7:05 a.m. to 12:10 p.m. = 5 h 5 min; subtracting the 20-minute stop gives 4 h 45 min of actual travel.
25. **C) 1 L 400 mL** — 4200 mL + 2800 mL = 7000 mL; 7000 ÷ 5 = 1400 mL per bottle.
26. **C) ₹195** — 9 × ₹24 = ₹216; ₹216 − ₹21 discount = ₹195.
27. **C) 900 L** — 1/9 × 1350 = 150 L used per day; over 3 days = 450 L used; 1350 − 450 = 900 L remaining.
28. **C) 49 kg** — 14200 g + 11800 g + 9300 g + 13700 g = 49000 g = 49 kg.
29. **C) 34** — 1224 ÷ 36 = 34 boxes.
30. **C) ₹300** — Sold for 60 × ₹45 = ₹2,700; profit = ₹2,700 − ₹2,400 = ₹300.

Section 4 — Achievers Section (2 marks each)

31. **C) 16** — Working backwards: (91 + 21) ÷ 7 = 112 ÷ 7 = 16.
32. **C) 24** — Let the first number be x. Then second = $3x$, third = $3x+12$. Sum: $x + 3x + (3x+12) = 180 \rightarrow 7x + 12 = 180 \rightarrow x = 24$.
33. **C) ₹1,920** — After spending 3/8 on books, 5/8 of the money remains. After spending 1/4 of that on stationery, 3/4 of 5/8 = 15/32 of the original remains. So 15/32 of the total = ₹900 → total = ₹1,920.
34. **D) 200** — Girls = (2 × 98) + 4 = 200. Check: 98 + 200 = 298. ✓
35. **C) 66** — Each term is the sum of the previous two: 25 + 41 = 66.

Scoring guide (out of 40, Level-2-difficulty paper): 28+ is a very strong result at this difficulty. 20–27 is solid. Below 20 — review *why* each missed question was harder (an extra step, a bigger number, two ideas combined) rather than the raw score.